Question ID: 167aff9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is $58 \text{ square centimeters } (\text{cm}^2)$, and the surface area of right rectangular prism Y is $1{,}450 \text{ cm}^2$. The volume of right rectangular prism Y is $1{,}250 \text{ cubic centimeters } (\text{cm}^3)$. What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

Correct Answer: 1260

Rationale

The correct answer is $1{,}260$. Since it's given that prisms X and Y are similar, all the linear measurements of prism Y are k times the respective linear measurements of prism X, where k is a positive constant. Therefore, the surface area of prism Y is k^2 times the surface area of prism X and the volume of prism Y is k^3 times the volume of prism X. It's given that the surface area of prism Y is $1{,}450 \text{ cm}^2$, and the surface area of prism X is 58 cm^2, which implies that $1{,}450 = 58k^2$. Dividing both sides of this equation by 58 yields $\frac{1{,}450}{58} = k^2$, or $k^2 = 25$. Since k is a positive constant, $k = 5$. It's given that the volume of prism Y is $1{,}250 \text{ cm}^3$. Therefore, the volume of prism X is equal to $\frac{1{,}250}{k^3} \text{ cm}^3$, which is equivalent to $\frac{1{,}250}{5^3} \text{ cm}^3$, or 10 cm^3. Thus, the sum of the volumes, in cm^3, of the two prisms is $1{,}250 + 10$, or $1{,}260$.